JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON WEDNESDAY 21st JANUARY 2026) **TIME : 9:00 AM TO 12:00 NOON**

MATHEMATICS — TEST PAPER WITH ANSWER

SECTION-A

1. If the domain of the function

$f(x) = \cos^{-1}\left(\frac{2x-5}{11-3x}\right) + \sin^{-1}(2x^2 - 3x + 1)$ is the interval $[\alpha, \beta]$, then $\alpha + 2\beta$ is equal to :

(1) 1 (2) 3

(3) 5 (4) 2

2. The area of the region, inside the ellipse $x^2 + 4y^2=4$ and outside the region bounded by the curves $y = |x| - 1$ and $y = 1 - |x|$, is :

(1) $2(\pi - 1)$ (2) $2\pi - \frac{1}{2}$

(3) $3(\pi - 1)$ (4) $2\pi - 1$

3. The number of relations, defined on the set $\{a, b, c, d\}$, which are both reflexive and symmetric, is equal to:

(1) 256 (2) 16

(3) 1024 (4) 64

4. Let a point A lie between the parallel lines L_1 and L_2 such that its distances from L_1 and L_2 are 6 and 3 units, respectively. Then the area (in sq. units) of the equilateral triangle ABC, where the points B and C lie on the lines L_1 and L_2 respectively, is :

(1) $15\sqrt{6}$ (2) 27

(3) $21\sqrt{3}$ (4) $12\sqrt{2}$

5. Let $\vec{a} = -\hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{b} = 8\hat{i} + 7\hat{j} - 3\hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} \times \vec{c} = \vec{b}$. If $\vec{c}.(\hat{i} + \hat{j} + \hat{k}) = 4$, then $|\vec{a} + \vec{c}|^2$ is equal to :

(1) 33 (2) 30

(3) 35 (4) 27

6. Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive terms such that $a_2.a_3.a_4 = 64$ and $a_1 + a_3 + a_5 = \frac{813}{7}$.

Then $a_3 + a_5 + a_7$ is equal to :

(1) 3256 (2) 3252

(3) 3244 (4) 3248

7. Let $\vec{c}$ and $\vec{d}$ be vectors such that $|\vec{c} + \vec{d}| = \sqrt{29}$ and $\vec{c} \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = (2\hat{i} + 3\hat{j} + 4\hat{k}) \times \vec{d}$. If $\lambda_1, \lambda_2 (\lambda_1 > \lambda_2)$ are the possible values of $(\vec{c} + \vec{d}).(-7\hat{i} + 2\hat{j} + 3\hat{k})$, then the equation

$$K^2x^2 + (K^2 - 5K + \lambda_1)xy + \left(3K + \frac{\lambda_2}{2}\right)y^2 - 8x + 12y + \lambda_2 = 0$$

represents a circle, for k equal to :

(1) 4 (2) 1

(3) –1 (4) 2

8. Let $y = y(x)$ be the solution curve of the differential equation $(1 + x^2)dy + (y - \tan^{-1}x)dx = 0$, $y(0) = 1$. Then the value of $y(1)$ is :

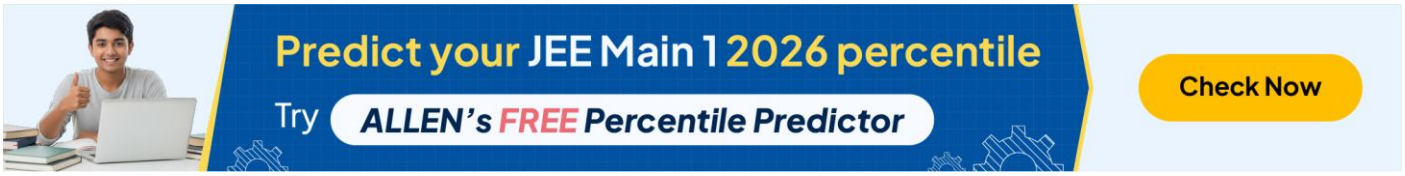
(1) $\frac{2}{e^{\frac{\pi}{4}}} + \frac{\pi}{4} - 1$ (2) $\frac{2}{e^{\frac{\pi}{4}}} - \frac{\pi}{4} - 1$

(3) $\frac{4}{e^{\frac{\pi}{4}}} + \frac{\pi}{2} - 1$ (4) $\frac{4}{e^{\frac{\pi}{4}}} - \frac{\pi}{2} - 1$

9. The number of strictly increasing functions f from the set $\{1, 2, 3, 4, 5, 6\}$ to the set $(1, 2, 3, \ldots, 9)$ such that $f(i) \neq i$ for $1 \leq i \leq 6$, is equal to :

(1) 21 (2) 27

(3) 22 (4) 28



10. Let $f : R \to (0, \infty)$ be a twice differentiable function such that $f(3) = 18$, $f'(3) = 0$ and $f''(3) = 4$.

Then $\lim\limits_{x\to 1}\left(\log_e\left(\frac{f(2+x)}{f(3)}\right)^{\frac{18}{(x-1)^2}}\right)$ is equal to :

(1) 1 (2) 9

(3) 2 (4) 18

11. Let the foci of hyperbola coincide with the foci of the ellipse $\frac{x^2}{36}+\frac{y^2}{16} = 1$. If the eccentricity of the hyperbola is 5, then the length of its latus rectum is:

(1) 12 (2) 16

(3) $\frac{96}{\sqrt{5}}$ (4) $24\sqrt{5}$

12. The value of $\int\limits_{-\pi/6}^{\pi/6}\left(\frac{\pi+4x^{11}}{1-\sin(|x|+\pi/6)}\right)dx$ is equal to

(1) 2π (2) 4π

(3) 8π (4) 6π

13. Let the mean and variance of 7 observations 2, 4, 10, x, 12, 14, y, x > y, be 8 and 16 respectively. Two numbers are chosen from {1, 2, 3, x–4, y, 5} one after another without replacement, then the probability, that the smaller number among the two chosen numbers is less than 4, is:

(1) $\frac{3}{5}$ (2) $\frac{4}{5}$

(3) $\frac{2}{5}$ (4) $\frac{1}{3}$

14. Let (α, β, γ) be the co-ordinates of the foot of the perpendicular drawn from the point (5, 4, 2) on the line $\vec{r} = (-\hat{i}+3\hat{j}+\hat{k})+\lambda(2\hat{i}+3\hat{j}-\hat{k})$.

Then the length of the projection of the vector $\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$ on the vector $6\hat{i}+2\hat{j}+3\hat{k}$ is :

(1) $\frac{15}{7}$ (2) 4

(3) $\frac{18}{7}$ (4) 3

15. Let PQ and MN be two straight lines touching the circle $x^2 + y^2 – 4x – 6y – 3 = 0$ at the points A and B respectively. Let O be the centre of the circle and $\angle AOB = \pi/3$. Then the locus of the point of intersection of the lines PQ and MN is:

(1) $3(x^2 + y^2) – 18x – 12y + 25 = 0$

(2) $x^2 + y^2 – 12x – 18y – 25 = 0$

(3) $x^2 + y^2 – 18x – 12y – 25 = 0$

(4) $3(x^2 + y^2) – 12x – 18y – 25 = 0$

16. If the coefficient of x in the expansion of $(ax^2 + bx + c)(1 – 2x)^{26}$ is – 56 and the coefficients of x^2 and x^3 are both zero, then a + b + c is equal to

(1) 1300 (2) 1500

(3) 1403 (4) 1483

17. If $x^2 + x + 1 = 0$, then the value of $\left(x+\frac{1}{x}\right)^4+\left(x^2+\frac{1}{x^2}\right)^4+\left(x^3+\frac{1}{x^3}\right)^4+\ldots+\left(x^{25}+\frac{1}{x^{25}}\right)^4$ is :

(1) 128 (2) 162

(3) 175 (4) 145

18. The value of $\text{cosec}10° - \sqrt{3}\sec 10°$ is equal to:

(1) 4 (2) 2

(3) 8 (4) 6



19. The sum of all the roots of the equation $(x-1)^2 - 5|x-1| + 6 = 0$, is:

(1) 4 (2) 3

(3) 1 (4) 5

20. Let O be the vertex of the parabola $x^2 = 4y$ and Q be any point on it. Let the locus of the point P, which divides the line segment OQ internally in the ratio 2: 3 be the conic C. Then the equation of the chord of C, which is bisected at the point (1, 2), is:

(1) $5x-y-3=0$ (2) $4x-5y+6=0$

(4) $x-2y+3=0$ (4) $5x-4y+3=0$

SECTION-B

21. Let $f : R \to R$ be a twice differentiable function such that the quadratic equation $f(x)m^2 - 2f'(x)m + f''(x)=0$ in m, has two equal roots for every $x \in R$. If $f(0)=1$, $f'(0) = 2$ and (α, β) is the largest interval in which the function $f(\log_e x - x)$ is increasing, then $\alpha + \beta$ is equal to

22. Let $a_1 = 1$ and for $n \geq 1$, $a_{n+1} = \frac{1}{2}a_n + \frac{n^2-2n-1}{n^2(n+1)^2}$. Then $\left|\sum_{n=1}^{\infty}\left(a_n - \frac{2}{n^2}\right)\right|$ is equal to _____.

23. Let $S=\{(m, n): m, n \in \{1, 2, 3, ..., 50\}\}$. If the number of elements (m, n) in S such that $6^m + 9^n$ is a multiple of 5 is p and the number of elements (m, n) in S such that m + n is a square of a prime number is q, then p + q is equal to……..

24. For some $\alpha, \beta \in R$, let $A = \begin{bmatrix} \alpha & 2 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & \beta \end{bmatrix}$ be such that $A^2 - 4A + 2I = B^2 - 3B + I = O$. Then $(\det(\text{adj}(A^3 - B^3)))^2$ is equal to ……

25. $6\int_0^{\pi}|(\sin 3x + \sin 2x + \sin x)|dx$ is equal to……..



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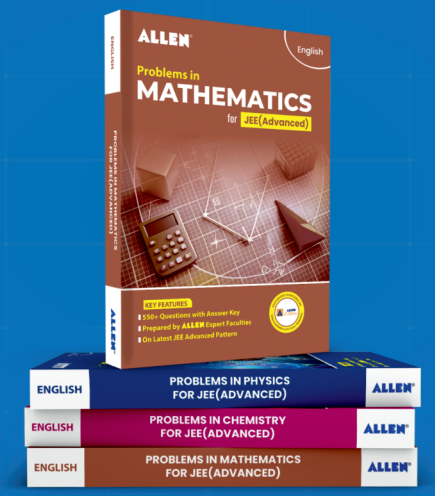


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